

A Low-Noise Design Technique for High-Speed CMOS Optical Receivers

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Abstract—A careful comparison between alternative topologies to realize low-noise wideband TIAs is carried out in this work. In order to break the tradeoff between noise and bandwidth, the proposed front-end uses two stages, i.e. a low-noise narrowband transimpedance interface followed by an equalizer aimed at restoring the required bandwidth. The technique is especially effective for white noise components. The core first-stage amplifier exploits current reuse for minimum power consumption and is optimized for colored noise reduction. A net $4\times$ noise power reduction is achieved if compared with a design approach based on a traditional shunt-feedback TIA with the same bandwidth. A complete receiver, interfacing a commercial photodiode, and including the proposed two-stage front-end (TSFE), a limiting amplifier and a wideband output buffer has been realized in 65 nm CMOS. Optical communications tailored to 100GBASE-LR4 standard, which is specified for mid-to-long range transmissions at a channel rate of 25 Gb/s, are targeted. Realized prototypes show a sensitivity of -11.9 dBm at a BER of 10^{-12} with a PRBS31 input pattern and a transimpedance gain of 83 dB Ω , while tolerating an overall input capacitance of 160 fF. To the best of the authors' knowledge, this is the best sensitivity performance achieved by 25-Gb/s optical receivers in CMOS, comparable to state-of-the-art BiCMOS realizations.

Index Terms—CMOS technology, current reuse, equalization, input-referred noise, optical receivers, shunt-feedback, transimpedance amplifiers (TIA).

I. INTRODUCTION

THE fast-growing bandwidth needs from the Internet, super-computing, and data centers require ever increasing data rates, where electrical communications media not only have fairly limited reach, but also are less efficient, leading to high and eventually unaffordable power consumptions [1]–[4]. Optical communications, once served mainly for telecommunications backbone networks, are expanding territory to lower communication hierarchy and penetrating lower span, e.g., board-to-board, chip-to-chip, and eventually intra-chip. This trend is further accelerated by recent technology advancements on silicon photonics, which enables optical devices fabricated on CMOS platforms for mass volume [5], [6].

Manuscript received September 17, 2013; revised January 24, 2014; accepted April 28, 2014. Date of publication May 19, 2014; date of current version May 28, 2014. This paper was approved by Associate Editor Jared Zerbe.

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Digital Object Identifier 10.1109/JSSC.2014.2322868

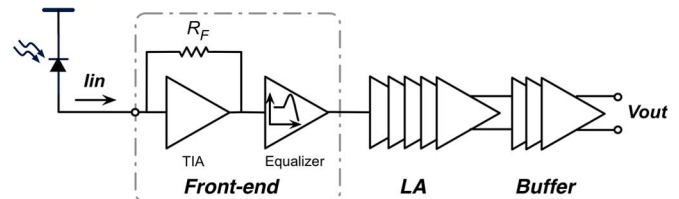


Fig. 1. Receiver block diagram.

In this work, we describe the design and characterization of an optical receiver tailored to IEEE 100GBASE-LR4 standard for mid-to-long-range transmissions at a channel speed of 25 Gb/s [7]. The IC interfacing an off-chip commercial photodiode comprises a two-stage front-end (TSFE), cascading a transimpedance amplifier (TIA) and an equalizer, followed by a five-stage limiting amplifier (LA) and a buffer, as shown in Fig. 1.

TIAs usually limit the receiver noise, and solutions beyond 10 Gb/s with noise performance adequate for the applications are challenging. Common-gate (CG) stages are known for their superior high-frequency operation but they show unfavorable noise [12]. A modified approach might alleviate the noise problem but at the price of power consumption [9]. The shunt-feedback (SF)-based TIA, traditionally considered a low-noise topology, is losing advantage in a high-data-rate scenario, where the bandwidth requirement sets a limit to the maximum feedback resistance value, thus determining a minimum noise level for the TIA. As an example, the SF TIA shown in [10], targeting short-range multimode fiber at 850 nm, displays an input-referred noise current of $4.2 \mu\text{A}_{\text{rms}}$, unsuitable for mid-to-long-range optical transmission like 100GBASE-LR4, where lower noise levels are generally required. Some conventional optimization techniques can be applied to the two basic alternatives so as to alleviate the noise issue. In [8], capacitive-matching technique [11], [12] is used, but still noise is unfavorable. Input series peaking is also used to cancel portions of the capacitance and lower noise at high frequency [13]–[15]. However, these techniques help at best in a moderate way. Instead, we propose a low-noise two-stage front-end (TSFE), realized by the series combination of a narrowband TIA followed by an equalizer aimed at restoring the application bandwidth [16]. The proposed TSFE proves effective for realization in low supply technologies and achieves low-noise performance. Prototypes of the optical receiver have been realized in 65 nm CMOS. Experiments show sensitivity of -11.9 dBm at a BER of 10^{-12} with a PRBS31 input pattern and

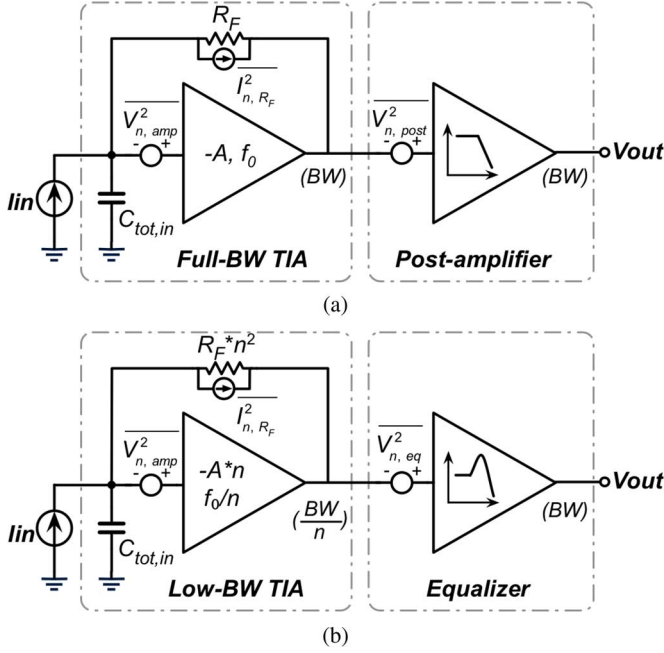


Fig. 2. Front-end topologies. (a) SF TIA. (b) TSFE.

a transimpedance gain of 83 dBΩ, while tolerating an overall input capacitance of 160 fF. A comparison with recent published works demonstrates the proposed solution goes beyond CMOS realizations [8]–[10], [22] in terms of sensitivity, and is comparable to state-of-the-art BiCMOS alternatives [24].

This paper is organized as follows. Section II provides insights into high data rate low-noise TIA topologies and introduces the TSFE. Section III gives the design details of the complete optical receiver. Section IV presents the experimental results, and Section V draws conclusions.

II. LOW-NOISE DESIGN TECHNIQUE FOR RECEIVER FRONT-END

We start reviewing the SF TIA, shown in Fig. 2(a), with noise sources annotated, identifying the main reasons for its limited noise performance at high data rate. A post-amplifier takes into account the main noise contribution from following stages. Then, we introduce and analyze the TSFE, shown in Fig. 2(b), comparing its noise performance versus the SF TIA alternative at full bandwidth.

A. Review of SF TIA

By inspection of the circuit, the input-referred noise power spectral density is derived as

$$\begin{aligned} \overline{I_{n,\text{in,SF}}^2} &= \overline{I_{n,R_F}^2}(s) + \overline{I_{n,\text{amp}}^2}(s) + \overline{I_{n,\text{post}}^2}(s) \\ &= \frac{4kT}{R_F} + \frac{4kT\gamma}{g_m R_F^2} |1 + sR_F C_{\text{tot,in}}|^2 + \frac{\frac{4kT\gamma}{g_{m,\text{post}}}}{|Z_T(s)|^2} \end{aligned} \quad (1)$$

where $C_{\text{tot,in}}$ is the total input capacitance at TIA input, comprised of photodiode capacitance C_{PD} and TIA input capacitance $C_{\text{in,tia}}$. $Z_T(s)$ is the TIA transimpedance transfer function. $\overline{V_{n,\text{amp}}^2} = 4kT\gamma/g_m$ and $\overline{V_{n,\text{post}}^2} = 4kT\gamma/g_{m,\text{post}}$ are

the equivalent input-referred noise power spectral densities of the core amplifier and the post-amplifier respectively. $\overline{I_{n,R_F}^2}$ is the thermal noise power spectral density of feedback resistor. g_m and $g_{m,\text{post}}$ are the transconductances of the input devices for core amplifier and post-amplifier and γ is the channel-noise factor. Equation (1) assumes a large amplifier gain A so that the noise contributions of amplifiers' loads can be disregarded and the transimpedance gain $|Z_T(s)|$ equals R_F at low frequency.

In low-frequency applications, SF TIA is usually modeled as a first-order system where the input RC constant dominates the closed-loop response and the dominant pole f_0 of core amplifier is sufficiently high to be neglected. From (1), a large R_F minimizes the input-referred noise power spectral density, not only suppressing feedback resistor noise, but also the white noise components from both amplifier stages. On the other hand, increasing R_F requires increasing the amplifier gain A in direct proportion in order to recover the original bandwidth $BW = (A + 1)/(2\pi R_F C_{\text{tot,in}})$. Unfortunately, higher gain at higher frequency are contrasting requirements, setting an upper R_F value for given rate and bandwidth.

In fact, as data rate increases, the first-order SF model is furthermore insufficient to characterize the circuit since the pole f_0 of core amplifier starts limiting the bandwidth. In this case, considering a second-order closed-loop behavior, to have adequate phase margin the maximum feedback resistance is bounded. The so-called transimpedance limit can be derived as [12], [17]

$$R_F \leq \frac{A * f_0}{2\pi C_{\text{tot,in}} BW^2}. \quad (2)$$

Because $A * f_0$, corresponding to the gain–bandwidth product of the core amplifier, is a technology-dependent parameter, R_F trades with the square of TIA bandwidth if the total input capacitance does not change. The transimpedance limit sets a rapid growth of noise in high-data-rate applications. Let us assume the data rate rises by n times (bandwidth correspondingly scales up n times too). From (2), R_F becomes n^2 times smaller, and from (1) R_F noise boosts n^2 times and the white noise of amplifiers n^4 times. Therefore, the white noise from R_F and amplifiers grow much faster than bandwidth increase, rendering SF topology less appealing in high data rate applications.

B. Low-Noise TSFE

Decoupling noise and bandwidth requirements through a TSFE, comprised of a narrow-bandwidth SF TIA followed by an analog equalizer restoring the required bandwidth, can be an effective idea to achieve low-noise performance at increasing bandwidth. The block diagram of the proposed solution is reported in Fig. 2(b). In order to assure the same shape of frequency response (the same Q factor of closed-loop poles) as for SF TIA, the amplifier gain A needs to scale up n times while the pole f_0 down n times, as discussed in the Appendix. The white noise advantage of a TSFE with a feedback resistor equal to $n^2 R_F$ versus R_F is net as discussed in Section II-A. Still, a closer insight is necessary in order to inspect colored noise within the bandwidth of interest. In fact, from (1), the zero in the input-referred noise power spectral density of core amplifier is moved to lower frequency. Meanwhile, both poles of the TIA in open-loop need to be taken into account when

deriving Z_T with the consequence of an increased contribution from equalizer noise with increasing frequency.

Starting with the classical SF TIA, since the closed-loop response is usually designed with Butterworth shape¹ [12] for maximum flat gain without gain peaking, the TIA frequency response is given by

$$|Z_T(s)|^2 \approx \frac{R_F^2}{\left|1 + \frac{s}{2\pi BW}\right|^4}. \quad (3)$$

By means of (3) and (1), we derive (4) expressing input-referred noise power spectral density for the SF TIA, with a feedback resistor equal to R_F as follows:

$$\begin{aligned} \overline{I_{n,\text{in,SF}}^2}(f) = & \frac{4kT}{R_F} + \frac{4kT\gamma}{g_m R_F^2} + 4kT\gamma \frac{(2\pi C_{\text{tot,in}})^2}{g_m} f^2 \\ & + \frac{4kT\gamma}{g_{m,\text{post}} R_F^2} + \frac{4kT\gamma}{g_{m,\text{post}} R_F^2} \left(\frac{f}{BW}\right)^4 \end{aligned} \quad (4)$$

while the following equation reports the TSFE input-referred noise power spectral density assuming $n^2 R_F$ as feedback resistor and same overall bandwidth, i.e., same rate as SF TIA:

$$\begin{aligned} \overline{I_{n,\text{in,TSFE}}^2}(f) = & \frac{4kT}{R_F n^2} + \frac{4kT\gamma}{g_m R_F^2 n^4} \\ & + 4kT\gamma \frac{(2\pi R_F n^2 C_{\text{tot,in}})^2}{g_m R_F^2 n^4} f^2 \\ & + \frac{4kT\gamma}{g_{m,\text{eq}} R_F^2 n^4} + \frac{4kT\gamma}{g_{m,\text{eq}} R_F^2 n^4} \left(\frac{f}{BW/n}\right)^4 \\ = & \frac{4kT}{R_F n^2} + \frac{4kT\gamma}{g_m R_F^2 n^4} + 4kT\gamma \frac{(2\pi C_{\text{tot,in}})^2}{g_m} f^2 \\ & + \frac{4kT\gamma}{g_{m,\text{eq}} R_F^2 n^4} + \frac{4kT\gamma}{g_{m,\text{eq}} R_F^2 n^4} \left(\frac{f}{BW}\right)^4 \end{aligned} \quad (5)$$

where $g_{m,\text{eq}}$ stands for the transconductance of input transistor of the equalizer. Comparing (4) and (5), if $g_{m,\text{eq}} = g_{m,\text{post}}$, colored noise, namely f^2 and f^4 terms, is coincident in the two cases. This is made evident in Fig. 3 plotting different contributions of the two equations in logarithmic scale.

As far as the equalizer is able to recover a given bandwidth, the proposed strategy evidences a net noise advantage, because colored noise remains the same in the two cases whereas white noise significantly reduces. In practice, such an equalizer might be difficult to design when a considerable amount of bandwidth needs to be recovered and consequently n can't be arbitrarily large because: 1) the equalizer can only recover the bandwidth to a finite degree, and excessive scaling of the TIA bandwidth plus large amount of equalizer peaking tend to increase gain ripple and group delay variation, thus worsening ISI and reducing the SNR advantage coming from TSFE and 2) equalizer working at its limit will decrease the tunability, making the circuit susceptible to process, temperature and voltage (PVT) variations. Another limitation comes from the fact that the amplifier gain A needs to scale up n times at the same time, which is challenging to implement due to the low supply voltage in scaled

¹In this case, closed-loop (complex) poles quality factor equals $1/\sqrt{2}$ and the poles frequency equals the -3 dB bandwidth.

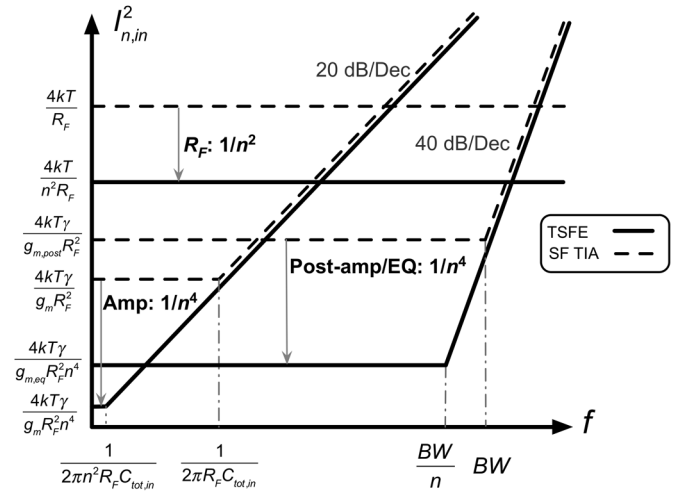


Fig. 3. Theoretical input-referred noise power spectral density (log scale): TSFE versus SF TIA.

CMOS. But noise power reduction is proportional to n^2 and n^4 , implying that even a moderate n (e.g. $n = 2$, $n^2 = 4$ and $n^4 = 16$) already provides sufficient noise suppression, leaving the rest of the white noise insignificant or even negligible.

III. RECEIVER DESIGN

Here, we describe the details in the implementation of the building blocks making up the proposed receiver, realized in 65 nm bulk CMOS technology.

A. TSFE

The receiver is intended to be connected to a commercial external photodiode through wire-bonding. Fig. 4 shows the schematic of the low-noise front-end. The photodiode provides single-ended input current, but the front-end still adopts a pseudo-differential structure employing a dummy mirror TIA to gain better common-mode noise suppression: the input current is fed to the main TIA while the input of the dummy mirror TIA is left open. This inevitably doubles the integrated input-referred noise power and power consumption of the TIA, but proves necessary in order to improve common-mode rejection, with respect to both externally coupled and internally generated interfering signals. In fact, even in a single slice demonstrator, the front-end processes small signals but the following stages, e.g., LA and Buffer, usually work in limiting mode and generate noisy power and ground rails shared with front-end. Power supply rejection ratio (PSRR) is even more relevant in a four-slice realization, as in 100GBASE-LR4 standard.

A dc-controlled current sink I_{sink} is designed to draw the average photocurrent generated by the photodiode, to maintain the operating point of TIA stable and remove the offset induced by the photocurrent. In this testchip version the control loop is not implemented on-chip and the control voltage is set externally.

A receiver bandwidth of around $25 \times 2/3 \approx 17$ GHz is targeted to achieve data rate of 25 Gb/s. In our design, applying TSFE topology with $n = 2$, the TIA bandwidth is scaled

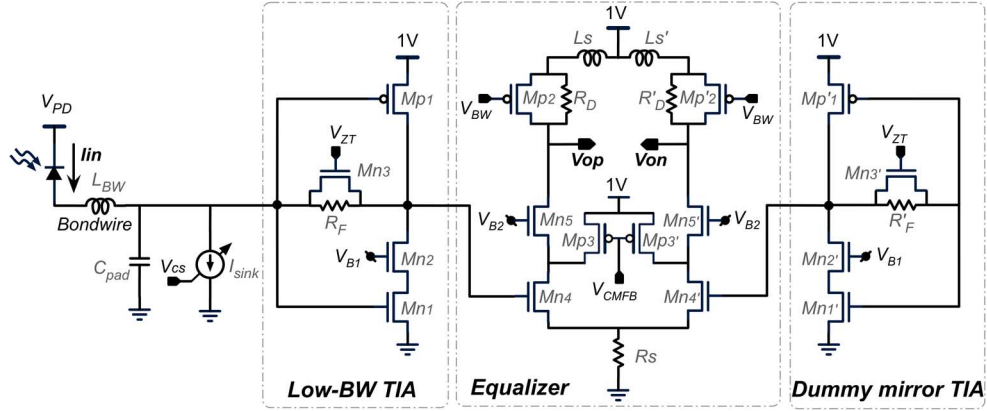


Fig. 4. Schematic of low-noise TSFE.

down by a factor of 2 to around 8.5 GHz. From (2), this enables $4\times$ higher R_F compared with a TIA of full bandwidth. Referring to Fig. 2(b), the amplifier needs also to increase its gain A and lower the pole f_0 by a factor of 2, as discussed in the Appendix. A CMOS inverter amplifier is thus adopted to provide the $2\times$ gain needed from SF TIA to TSFE, because it brings more gain than a NMOS based amplifier in low supply voltage thanks to current reuse [18]. An NMOS common-gate transistor Mn2 is inserted to lower the input capacitance, while increasing the output impedance and amplifier gain. Another PMOS cascode is not used due to limited voltage headroom. A CMOS amplifier has also net input-referred noise reduction by converting the PMOS load into a transconductance stage. On the other hand, the CMOS configuration brings along a higher input capacitance. From (4), the colored input-referred amplifier noise power (f^2 term), a quite significant noise component [12], has a coefficient of $C_{\text{tot},\text{in}}^2/g_m$. Thus, a proper ratio in the width of PMOS/NMOS is required to ensure that the benefit due to g_m increase is more than the increase of input capacitance $C_{\text{tot},\text{in}}$, so that $C_{\text{tot},\text{in}}^2/g_m$ is smaller and colored noise reduction is obtained. Detailed optimization is achieved through simulation setting the P/N ratio to 0.8/1 as a tradeoff between amplifier gain, bandwidth and noise. The noise reduction is about 40% compared with an NMOS amplifier assuming the same TIA bandwidth.

An extra NMOS Mn3 shunts R_F to make the feedback resistance tunable when a large current is fed in, to ensure proper TIA biasing in all operating conditions. When a small input current is injected, the NMOS is turned off, and the overall feedback resistance equals R_F .

Shunt peaking is employed in the equalizer, realized by means of a low-Q differential inductor $L_S/L_{S'}$ in series with a tunable resistance, made of a poly resistor R_D in parallel with a PMOS Mp2, controlled by V_{BW} . The low common-mode output voltage prevents the use of tail current for the differential pair and a poly resistor is used instead [15]. A cascode NMOS Mn5 is utilized to increase the output impedance of the transconductor and reduce the TIA load capacitance at the same time. Because the equalizer tuning through V_{BW} will alter the output common-mode voltage, a pair of PMOS active loads Mp3/Mp3' controlled by V_{CMFB} is inserted to steer current from the resistive load and to maintain the output

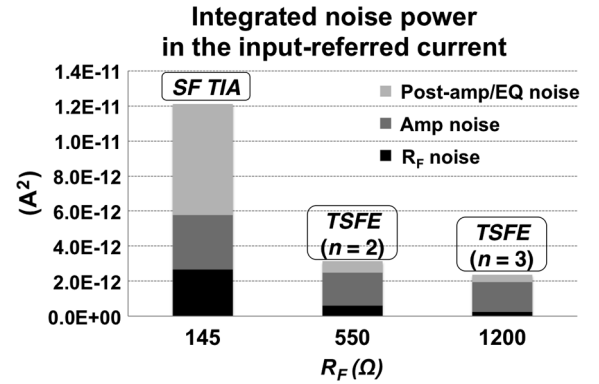


Fig. 5. Simulated noise reduction from SF TIA to TSFE.

common-mode voltage constant. The PMOS active load is connected to the drain of the transconductor instead of the output, so that the output capacitance is lowered. Overall, by tuning the variable resistance through control voltage V_{BW} , the equalizer can produce a peak variation from 0 dB to more than 10 dB, in order to add flexibility accommodating variations in parasitic capacitance and bondwire inductance.

In order to better assess the improvement of the proposed TSFE versus the conventional SF TIA approach, we have compared their noise performance through simulations using a single-ended version of the circuit shown in Fig. 4. In particular, we assume TSFE with $n = 2$ and $n = 3$. In all cases, the gain stage employs a CMOS structure with a classical P/N ratio of 2/1 for SF TIA [18], [19] and of 0.8/1 for TSFE. A cascode PMOS is also used for TSFE with $n = 3$ to achieve the required amplifier gain. R_F values are 145 Ω for SF, 550 Ω for TSFE $n = 2$, and 1200 Ω for TSFE $n = 3$, respectively. Total external input capacitance is 160 fF, assuming 80 fF for the photodiode and 80 fF contributed by the input pad. Since the external capacitance is dominant, the three TIAs have similar total input capacitance. Fig. 5 shows the integrated noise power in the input-referred current in the three cases. A $4\times$ total noise power reduction is observed from SF TIA to TSFE with $n = 2$. We also notice that the white noise of post-amplifier in SF topology is large because R_F is moderate. For TSFE with $n = 3$, the noise reduction becomes marginal because white noise components altogether no longer dominate and colored

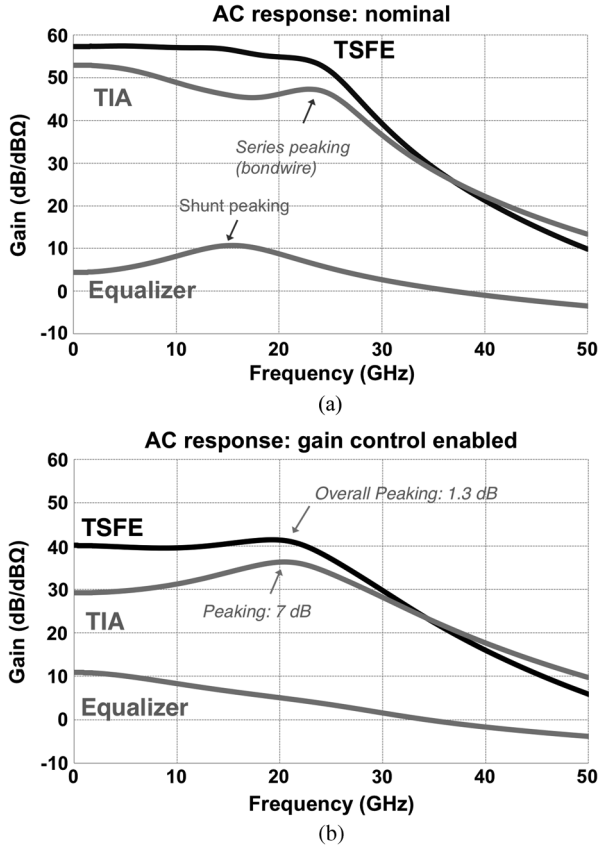


Fig. 6. Simulated ac response of TSFE. (a) Nominal configuration. (b) Minimum gain condition.

noise does not scale. Furthermore, since the bandwidth needs to be recovered from 5.7 GHz ($\approx 17 \text{ GHz} \div 3$) to 17 GHz in this case, the equalizer is tuned to its maximum peaking: not only the overall frequency response is not as flat as in the case of $n = 2$, but also the circuit has little tuning margin over PVT variation. The simulated group delay variation for SF is 6.6ps, while for TSFE with $n = 2$ and $n = 3$ it is 12.5 ps and 26.8 ps respectively. Eye diagram simulation with a $2^7 - 1$ nonreturn-to-zero pseudorandom bit sequence (PRBS7) shows data-dependent jitter of 1, 1.1, and 3.3 ps, respectively, for the three cases. The larger jitter in the last case is mainly due to gain ripple and larger group delay variation. We have thus selected $n = 2$ in the design of our TSFE. Circuit optimization considering the layout parasitics sets R_F to 510 Ω in the actual implementation.

Post-layout frequency response of TSFE is plotted in Fig. 6(a), assuming the effect of 500 μm -long bondwire, aimed at bandwidth enhancement. The series peaking frequency due to bondwire-photodiode connection is given by $f_{\text{peak}} \approx 1/(2\pi\sqrt{C_{\text{PD}}L_{\text{BW}}})$. For a C_{PD} of 80 fF and a bondwire inductance L_{BW} of 0.5 nH, peaking is at 24 GHz. The TIA stage alone has a bandwidth of 8.1 GHz. The equalizer boosts at high frequency due to shunt peaking at around 15 GHz. The TSFE provides overall 57.2 dB Ω transimpedance gain and 20.5 GHz bandwidth. TSFE output eye diagram with peak-to-peak amplitude set to sensitivity is plotted in Fig. 7. The input current is a PRBS7 with rise and fall time (10% $\sim$

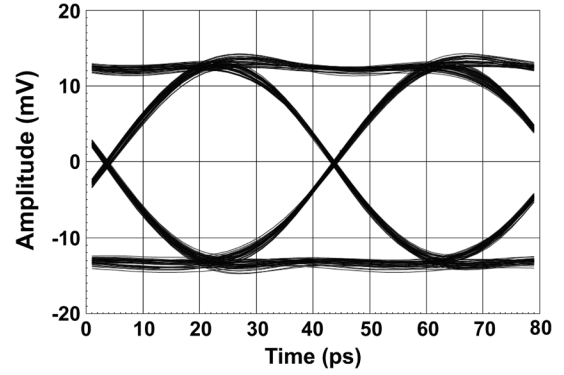


Fig. 7. Simulated output eye of TSFE with input current set to sensitivity (PRBS7 input).

90%) of 8.3 ps. The data-dependent jitter is less than 1.7 ps in this case, where the extra 0.6 ps jitter penalty versus previous analysis comes from post-layout parasitics and inclusion of series inductor. A PRBS7 is used in simulation since the time required to run a circuit level simulation using a PRBS31 is prohibitive.

The overload capability of TSFE is also a concern because a large input current, e.g. in the range of 2 mA_{pp}, would drive TIA into heavily nonlinear region. Furthermore, since the output common-mode voltage of TSFE is set to 800 mV, a large current may generate highly asymmetrical output swings with respect to common mode, which is especially unfavorable here since the equalizer stage is not fully differential. Consequently, large distortion and jitter would degrade the output eye significantly. However, gain control mechanism is intrinsic in TSFE architecture thanks to variable R_F and programmable equalizer, and gain control is thus applied to TSFE to alleviate the eye distortion at large input current. R_F can be made very small ($\sim 30 \Omega$) through V_{ZT} , which inevitably gives rise to considerable peaking of TIA stage. The equalizer then is tuned to a low-pass filter (-3 dB bandwidth of 10 GHz) to suppress the TIA peaking. The frequency response for the lowest achievable gain is plotted in Fig. 6(b).

B. Limiting Amplifier and Buffer

A five-stage limiting amplifier (LA) has been used to obtain gain-bandwidth product boost under reasonable power consumption and noise performance. The topology the LA is reported in Fig. 8. Inter-stage active-feedback is adopted to provide further bandwidth enhancement similar to [20] where the active-feedback is applied to single stage. The larger the active-feedback transconductance G_{mf} , the larger the overall bandwidth and smaller total gain. The transconductors of the last two stages (G_{m4} and G_{m5} within the active-feedback loop) are half-sized to speed up the forward path.

The LA gain stage schematic is plotted in Fig. 9(a). Shunt peaking is utilized to extend bandwidth. PMOS (Mp1/Mp1') operating in linear region constitutes the load resistance and can be tuned externally for optimum gain shape. A CMFB loop is added for two purposes: 1) regulating the output common-mode voltage over PVT variations; 2) feed current into the input differential pair devices to ensure high g_m , while not causing

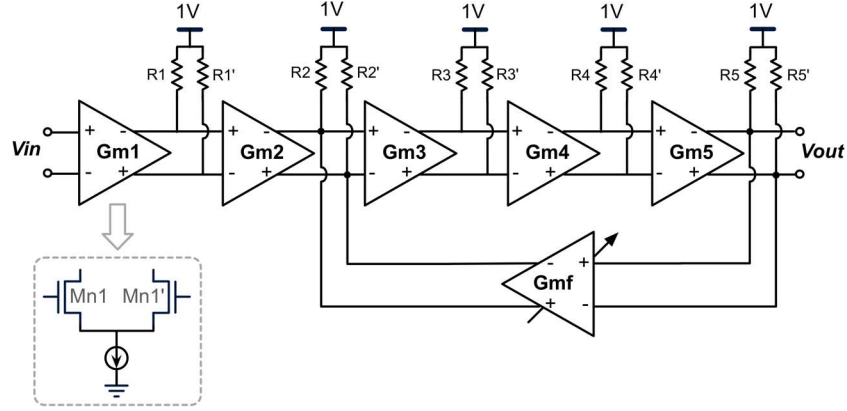


Fig. 8. Topology of LA.

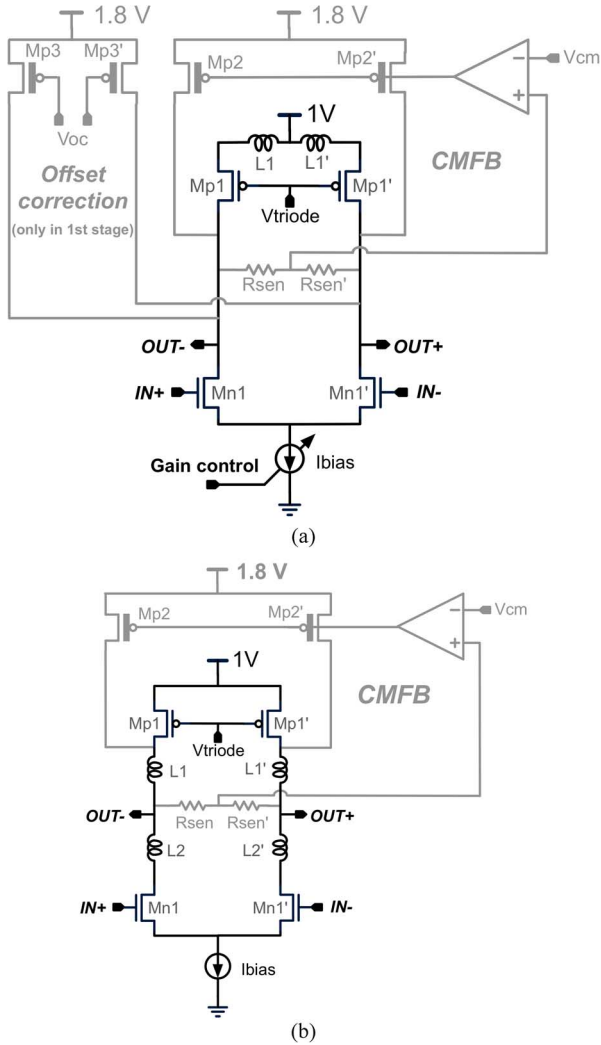


Fig. 9. Gain stage. (a) LA. (b) Output buffer.

voltage headroom issue when all current flows through the main load $Mp1/Mp1'$. The active loads $Mp2/Mp2'$ are realized by thick-oxide PMOS powered by a higher VDD (1.8 V) in favor of their large output impedance that causes smaller impact over the gain of LA stage. Overall, the LA achieves 23.2 dB gain and 19.4 GHz bandwidth.

The bias current of LA is programmable for gain control. When addressing large input voltage swings, the gain is lower (~ 11 dB) to avoid signal distortion and excessive data-dependent jitter. The CMFB loop regulates the output common-mode voltage in this tuning process. Offset correction is done in the first stage of LA through a pair of externally controlled thick-oxide PMOS active loads $Mp3/Mp3'$.

The output buffer is realized through three cascaded stages, as shown in Fig. 1, in a tapered sizing approach: the input stage size is small not to load LA while the output stage is sized large with on-chip termination to drive $50\ \Omega$ off-chip load. The gain stage schematic is drawn in Fig. 9(b) and resembles LA gain stage. In the output stage, $50\ \Omega$ poly-resistor replaces the linear PMOS $Mp1/Mp1'$ shown in Fig. 9(b) for better output matching in large-signal operation.

The output buffer uses a stagger-tuning approach, where the three stages provide gain peaking at 25, 43, and 30 GHz, respectively, by properly setting the values of $L1/L1'$ peaking inductors, and by adding an extra pair of inductors ($L2/L2'$). The buffer synthesizes an overall bandwidth of more than 50 GHz in nominal condition. PVT simulations show that in the worst case the buffer bandwidth is broader than 38 GHz and the gain ripple is less than 2 dB.

IV. EXPERIMENTAL RESULTS

The receiver chip has been fabricated by STMicroelectronics. The photomicrograph is shown in Fig. 10. The chip is pad limited and the core occupies $0.64\text{ mm} \times 0.66\text{ mm}$ (0.42 mm^2) including the I/O RF pads. The input pads are placed according to B-G-S-G-B order, where G-S-G is used for electrical probe characterization while the B-S-B is used to wire-bond the commercial photodiode for optical measurement. In such a way, both electrical and optical characterization is performed through the same chip. For optical measurements, photodiode bias voltage is provided via B pads. RF filters have been placed between B signal and G signal both off-chip and on-chip for good ground coupling. The input pad capacitance is 80 fF due to large pad dimension adopted for the ease of bonding procedure. The output pads are differential G-S-G-S-G, since the receiver has differential outputs.

The receiver draws 26.5 mA from 1 V and 36.9 mA from 1.8 V, resulting in a total dc power consumption of 93 mW.

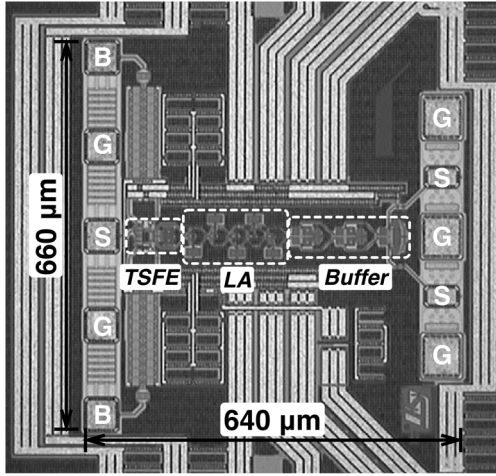
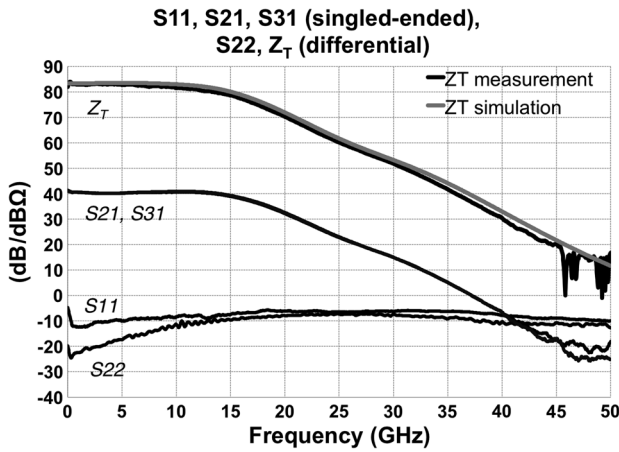


Fig. 10. Chip photomicrograph.

Fig. 11. Electrical characterizations of S - and Z -parameters: simulation and measurements.

Electrical characterization has been performed in order to compare simulations and measurements. S -parameter measurements have been carried out using a vector network analyzer (VNA) with 50 GHz bandwidth, by directly accessing the I/O pads via RF probes. Three-port characterization is applied to the receiver: one for the input and two for the outputs based on the “single-ended in, differential out” receiver I/O characteristics. Fig. 11 shows the resulting differential transimpedance gain, derived from measured S -parameters as [21]

$$Z_T = \frac{Z_0 (S_{21} - S_{31})}{1 - S_{11}} \quad (6)$$

where Z_0 is 50 Ω . A differential transimpedance gain of 83 dB Ω over a -3 dB bandwidth of 13.6 GHz has been measured. Comparison between simulations and measurements shows a very good agreement. Control of V_{BW} at the equalizer stage in the front-end allows tuning 10.6 GHz $\sim$ 18.2 GHz bandwidth and 87.1 dB Ω $\sim$ 78 dB Ω gain, respectively. Fig. 12 shows comparison between measured and simulated receiver group delay showing good agreement. In-band group delay variation is always below 20 ps. The curve is noisy because group delay is derived from sampled phase data.

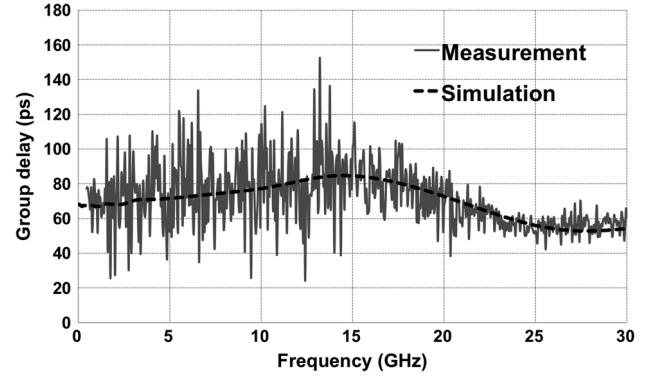


Fig. 12. Simulated and measured receiver group delay.

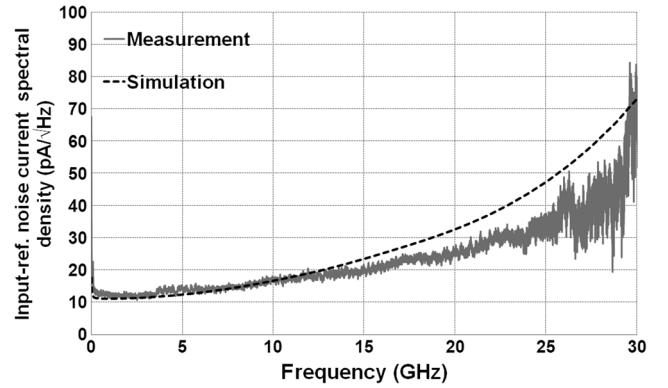


Fig. 13. Input-referred noise current spectral density: measurement versus simulation.

Measured and simulated input-referred noise current spectral density is reported in Fig. 13. The measurement has been performed by detecting the output noise voltage spectral density through Agilent N9030A spectrum analyzer with receiver inputs floating. Insertion loss from cable and Bias-Tee has been properly de-embedded. The outcome is then divided by the measured transimpedance gain of Fig. 11. The input-referred rms noise current has been computed as the measured output RMS noise (through integration of spectral density), divided by the measured in-band transimpedance gain, leading to 1.79 μ Arms. Dividing the input-referred RMS noise current by $\sqrt{BW}$ [12] determines an average input-referred noise current density of 15.3 pA/ $\sqrt{Hz}$. The noise measured is smaller than our previous result in [16] mainly due to more careful instrument setup.

Extensive optical measurements have been performed, including output eye diagram and bit error rate (BER) using $2^{31} - 1$ non-return-to-zero pseudorandom bit sequence (PRBS31). The measurement setup is shown in Fig. 14. The optical source is an Agilent 81672B tunable laser source with its wavelength set to 1310 nm. An Anritsu MP1800A signal quality analyzer is used for pattern generation and error detection. A Photline MXAN1300-LN-40 Mach-Zehnder modulator, driven by a Photline DR-DG-40-MO wideband driver amplifier, is adopted to generate receiver optical input signal. At the measurement biasing point of the modulator, 12.4 dB extinction ratio (ER) is obtained. The rise and fall times of the generated optical PRBS31 signal are around 9 ps,

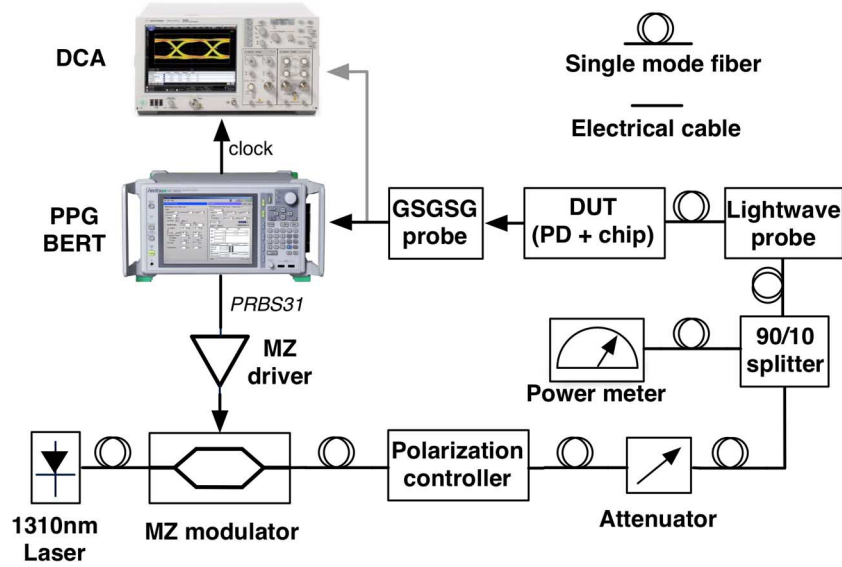


Fig. 14. Setup for optical eye diagram and BER test measurements.

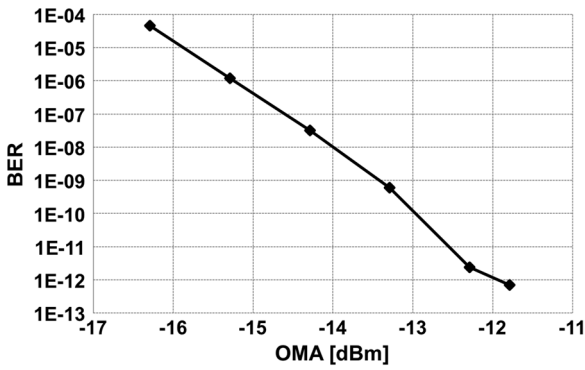


Fig. 15. Measured BER versus OMA input power at 25 Gb/s for a PRBS31 input pattern.

and the RMS jitter is 1.1 ps. A Cosemi LX3053 InGaAs-InP PIN photodiode with input capacitance of 80 fF is wire-bonded to the electrical chip to detect the light. Combining another 80 fF contributed by the input pad, the overall capacitance at receiver input is approximately 160 fF. Since the anode is dc connected to TIA input with a common-mode voltage of 430 mV, and the cathode is biased at 2.5 V, the reverse-bias voltage of photodiode is about 2.1 V and the measured photodiode responsivity is 0.91 A/W. An Agilent DCA-X 86100D sampling oscilloscope is used for eye diagram inspection.

The measured receiver BER with PRBS31 input data at different optical input power [optical modulation amplitude (OMA)] is shown in Fig. 15, demonstrating a receiver sensitivity of -11.9 dBm for BER of 10^{-12} , meeting the IEEE 100GBASE-LR4 sensitivity requirement of -8.6 dBm OMA [7] with significant margin.

The measured 25-Gb/s optical eye diagram at differential outputs for input powers at sensitivity and close to overload limit are shown in Fig. 16(a) and (b), respectively. At sensitivity level of -11.9 dBm OMA, the receiver output eye amplitude is 538 mV_{ppd} as shown in Fig. 16(a), sufficient to drive subsequent stages, e.g., CDR circuitry, in practical applications.

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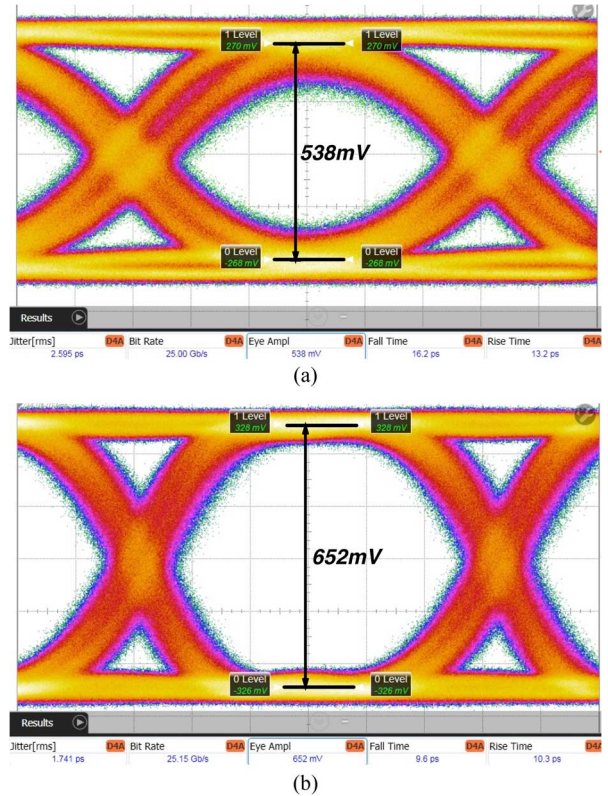


Fig. 16. Measured receiver output eye diagrams at 25 Gb/s for a PRBS31 input pattern at (a) sensitivity of -11.9 dBm OMA and (b) input power of 2.5 dBm OMA.

At large input signals, the 100GBASE-LR4 specification of 4.5 dBm for the maximum receive power (OMA) cannot be reached in our setup, due to the optical loss between the laser source and the DUT and lack of optical amplifier. Instead, the maximum optical power received by the photodiode is limited to 2.5 dBm OMA, which translates to input signal current of 1.6 mA_{pp}. By applying the gain control technique described

TABLE I
COMPARISON OF RECENT REPORTED 25-Gb/s OPTICAL RECEIVERS

Reference		[23]	[24]	[8]	[22]	[9]	[10]	This work
Technology		0.25 μ m-BiCMOS	0.13 μ m-BiCMOS	130nm SOI CMOS	90nm CMOS	65nm CMOS	65nm CMOS	65nm CMOS
Photodiode capacitance (fF)		220*	65	20	80	N/A	150	160*
Photodiode responsivity (A/W)		N/A	0.75	0.8	0.55	0.8	0.47	0.91
Transimpedance (dB Ω)		59	76.5	67	78.3	76.8	72.5	83
Power supply (V)		2.5	3.3	1.2	1.2	1.0 / 3.3	1.2	1.0 / 1.8
Power (mW)		28.2	67.5	48	44.4	137.5	69	93
Input-referred RMS noise current (μ Arms)		3.45	2.4	N/A	2.3	2.6	4.2	1.8
Sensitivity test for BER=10 ⁻¹² at 25Gb/s	PRBS pattern	15	31	N/A	7	9	7	31
	Sensitivity OMA (dBm)	N/A	-12	-6****	-4***	-9.7	-6.8	-11.9
	Modulation current (μ App)	93	47**	N/A	219**	86	98**	59

*: including input pad capacitance, **: calculated from reported data, ***: at 22Gb/s, ****: average power

in Section III-A, a wide open output eye is present, as shown in Fig. 16(b), reaching an output swing of 652 mV_{ppd}. On the contrary, without implementing gain control, this input signal level is already sufficient to heavily distort the output eye, as explained in previous section. The actual receiver overload limit is simulated to be 5.7 mA_{pp}, corresponding to 8 dBm OMA, when further turning on LA gain controls, and is limited by the maximum dc photocurrent that can be accommodated by the DC current sink I_{sink} within TSFE. The measured rms jitters of output eyes for nominal and large signal condition are 2.6 and 1.7 ps, respectively.

Table I compares this work with recent published 25-Gb/s optical receivers. This work achieves the lowest input-referred rms noise current, demonstrating the effectiveness of proposed low-noise design techniques. Compared with [10] which adopts the same technology and similar input capacitance, this work achieves a more than 5 dB better sensitivity. The low noise ensures the best reported receiver sensitivity at BER of 10⁻¹² for 25 Gb/s in CMOS technology, and comparable with state-of-the-art realization in 0.13 μ m-BiCMOS [24]. Note that the sensitivity is achieved in the worst-case test pattern, i.e., the PRBS31, in contrast to other CMOS designs reported in Table I using more test-friendly data patterns like PRBS7 and PRBS9. Furthermore, the high transimpedance gain ensures large output voltage swing of 538 mV_{ppd} even at sensitivity input level, i.e., sufficient to drive subsequent stages like CDR in a practical situation and more than 4 $\times$ larger than state-of-the-art BiCMOS [24].²

²Estimated from the reported eye diagram.

V. CONCLUSION

The design of low-noise receivers for optical communications entails a tradeoff between noise and bandwidth. The detailed analysis in this work demonstrates that the proposed strategy of a low-noise narrowband TIA followed by a bandwidth equalizer is effective and produces a significant noise reduction compared to a full bandwidth single stage TIA. In particular, the proposed solution designed in 65 nm CMOS for a 25-Gb/s rate achieves an optimum improvement when selecting a first stage with roughly half the bandwidth. The receiver testchip has been interfaced to a commercial photodiode and characterized both at sensitivity and at large input power levels, demonstrating a sensitivity performance significantly better than recently published 25-Gb/s CMOS optical receivers and comparable with state-of-the-art BiCMOS realizations, thus ensuring a safe optical margin with respect to the 100GBASE-LR4 standard specifications.

Overall, high sensitivity, large overload capability, and large output voltage swing achieved by this work demonstrates that optical receiver circuitry made in a mainstream CMOS is able to compete their SiGe or III-V counterparts from 10 Gb/s to a higher data rate of 25 Gb/s.

APPENDIX

For TIA assuming a single-pole core amplifier with large gain A shown in Fig. 2(a), the closed-loop transfer function is second-order, expressed as [12]

$$Z_T(s) \approx -\frac{R_F}{1 + \frac{s}{\omega_n Q} + \frac{s^2}{\omega_n^2}} \quad (\text{A1})$$

where the pole frequency ω_n is given by

$$\omega_n = \sqrt{\frac{A+1}{R_F C_{\text{tot},\text{in}}/(2\pi f_0)}} \quad (\text{A2})$$

and the Q factor is

$$Q = \frac{\sqrt{(A+1)R_F C_{\text{tot},\text{in}}/(2\pi f_0)}}{R_F C_{\text{tot},\text{in}} + 1/(2\pi f_0)} \approx \sqrt{\frac{A}{R_F C_{\text{tot},\text{in}} * 2\pi f_0}}. \quad (\text{A3})$$

For the TIA within TSFE in Fig. 2(b), assuming $C_{\text{tot},\text{in}}$ doesn't change and feedback resistance becomes $n^2 R_F$, to maintain the Q factor constant, the core amplifier gain A needs to scale up n times while the amplifier pole f_0 needs to scale down n times based on (A3), which is doable since the amplifier gain-bandwidth product does not change.

For Butterworth response, the TIA bandwidth is given by [12]

$$BW \approx \frac{\sqrt{2}A}{2\pi R_F C_{\text{tot},\text{in}}}. \quad (\text{A4})$$

Thus for TSFE the TIA bandwidth is scaled down by a factor of n , when feedback resistance scales up n^2 times and amplifier gain A scales up n times.

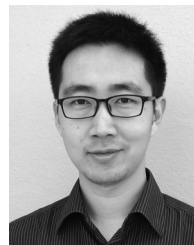
ACKNOWLEDGMENT

The authors would like to thank G. Giuliani and I. Cristiani from the University of Pavia for optical measurement support. This work has been carried out within the *Studio di Microelettronica*, a joint research laboratory between Università degli Studi di Pavia and STMicroelectronics.

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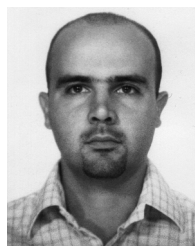
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